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Senior Software Engineer & Tech Lead

Software Architecture & AI Solutions · .NET · Python

Senior Software Engineering · Technical Leadership · Software Architecture · AI Solutions Engineering

Summary

Senior software engineer and Tech Lead with about 7 years of experience (since 2019). I design and deliver complete business applications in .NET and Python: architecture, APIs, web and mobile, AWS / Azure cloud and AI model integration.

After demanding industrial environments (**Schneider Electric**, SAFe trains of 90+ people) and **large-scale automation on AWS** (Ayming), I now own products end to end: **Tech Lead and architect at DREVIO**, architect and developer of my own solutions as a freelancer (ICEKERA).

Core skills:

- **Senior Software Engineering: complex business applications in .NET and Python, legacy modernisation, large-scale automation, BI**
- **Technical Leadership: mentoring developers, splitting work into deliverable packages, code review and quality of deliverables; certified SAFe Scrum Master**
- **Software Architecture**: designing complete solutions, technology choices, APIs and workflows, system integration, security
- **AI Solutions Engineering: local RAG, LLMs (Qwen, OpenAI, Claude), image analysis, agents and MCP servers, with evaluation and human oversight**

I integrate AI as a **first-class software component**, held to the same standards as the rest of the system: architecture, tests, data security, maintainability and **human oversight** of the results.

Stack C# .NET

Front	ASP.NET (Blazor), WPF
API	Authentication, Authorization, JWT, Middlewares
Caching	Output Caching, Redis
ORM	EntityFramework, PostgreSQL, SQL Server
Messaging	RabbitMQ
Logging	Serilog
Testing	xUnit, Moq, Playwright, K6, Test containers
Streaming	Apache Kafka
Real Time	SignalR
Task Scheduling	Background Service, PeriodicTimer
Cloud	Azure, AWS RDS, S3, DynamoDB
CI/CD Pipelines	GitHub Actions, Jenkins, Azure
Patterns	CQRS, Mediator, REPR
Architectures	Clean Architecture, Vertical Slice
Libraries	AutoMapper, MediatR, Fluent Validation, Swagger

Stack Python

Framework	Odoo
BackEnd	Django, FastAPI
Scraping	BeautifulSoup, Requests, Playwright
Data Analysis	Pandas, Numpy
ML	TensorFlow, Pytorch, NEAT, OpenCV
ETLs	CSV, XML, PDF

+ every area already covered by the .NET stack

Other

Front	JavaScript, TypeScript, React, (HTML, CSS), Sass/Scss, Tailwind
Orchestration	SnapLogic
DevOps	Docker, Jenkins, GitHub Actions, Azure DevOps, AWS ECS
AWS	ECS, RDS, DynamoDB, S3
Azure	Azure AD, Azure Blob Storage, Azure SQL
Security	JWT, OAuth2, AES-256, Secrets Management

AI stack

LLM	Qwen, OpenAI API, Claude API, Ollama
RAG	Qdrant, BM25, bge-m3, cross-encoder reranking
Vision & OCR	OpenAI (image analysis), OpenCV, RapidOCR
Agents	MCP, Claude Code, OpenClaw
Evaluation	Test sets, abstention thresholds, verified citations

Professional Experience

Independent Senior Software Engineer, Software Architect & AI Solutions, IT Consultant — Since 07/2026

ICEKERA - Freelance (El Yoel PEPIN)

Company context:

Freelance business under the ICEKERA brand: software architecture and development, artificial intelligence solutions, Tech Lead and IT consulting.

Missions:

1. ERP KERA - Management and invoicing ERP, ICEKERA product (10/2026 to present):

Management software that I design, build and sell to freelancers and small businesses in Guadeloupe (erp.kera.gp): quotes, invoices, CRM, finance, tax returns and e-invoicing. An AI assistant can work in it through a built-in MCP server, under the user's control.

➢ **Modular monolith architecture** on ASP.NET Core (.NET 10): 17 modules (sales, invoicing, CRM, finance, forecasting, projects...), enabled per company, with boundaries enforced by architecture tests.

➢ **Built-in MCP server**: an AI assistant (Claude Desktop, claude.ai, Claude Code) reads the data and prepares drafts; every sensitive action (sending, issuing, recording a payment) creates an approval request validated by the user in the ERP.

➢ **Securing AI access**: OAuth 2.1 (PKCE), Read or Drafts permissions, rate limits, data-sourced texts marked as untrusted (protection against prompt injection).

➢ **E-invoicing**: Factur-X, and URSSAF / VAT returns prepared from recorded payments.

➢ **React interface** with types generated from the API, light and dark themes, input protected against any loss (local backup copy).

➢ **Docker go-live** with database backup, migrations, health check and automatic rollback on failure.

➢ **Results:**

- 3,000+ automated tests: about 1,960 on the .NET side (including Testcontainers integration tests) and 1,185 on the interface.
- In production at erp.kera.gp since October 2026.

➢ **Tech environment:** C#, .NET 10, ASP.NET Core, Entity Framework Core, OpenIddict, React, TypeScript, Vite, Vitest, PostgreSQL, S3 storage, MCP (Model Context Protocol), OAuth 2.1, Docker, GitHub Actions, Testcontainers.

2. SESAM Interventions - On-site intervention survey app (offline PWA) (10/2026 to present):

Follow-up to the SESAM assignment: the site manager (schools and buildings of the Departmental Council) was writing his intervention surveys and cost breakdowns in Excel. The app replaces them, on the phone as well as at the office, even with no network on site.

➢ **Scoping** from the recording of the client meeting (Whisper transcription): needs and gaps with the app listed and prioritised, then analysis of the client's real Excel workbooks.

➢ **Offline PWA**: survey per site (rooms, equipment, photos), data stored on the phone (IndexedDB) and synchronised with the server when the network is back.

➢ **Geotagged photos** taken on site or picked from the gallery, date and GPS position read from the photo (EXIF).

➢ **Cost breakdown** computed on the phone: labour and equipment per contractor, margin, contingencies, subcontractor VAT, amounts excl. and incl. VAT, one quote number per craftsman.

➢ **Tracking table** based on the client's workbooks (one line per contractor, margin coefficient per site), attached quotes, invoices and delivery notes, cost-breakdown PDF ready to send.

➢ **Per-user accounts** (JWT); FastAPI / PostgreSQL API limited to authentication, synchronisation and photo storage.
➢ **Results:**
- In production since October 2026, deployed with Docker behind Traefik.

➢ **Tech environment:** React 19, TypeScript, Vite, PWA, Dexie (IndexedDB), jsPDF, FastAPI, SQLAlchemy, PostgreSQL, Docker, Traefik.

3. SESAM Video surveillance - Replacing SmartPSS (upcoming):
Next assignment for the same group: replace SmartPSS, the software currently used to view the company's cameras, with a solution that is easier to use day to day.
➢ **Viewing**: easier access to all cameras.
➢ **Mobile alerts** based on defined detection zones.
➢ **Connection** with the alert app the client already uses.
➢ **Existing setup:** IP cameras, SmartPSS (being replaced).

4. SESAM - Taking back control of and reorganising the IT system of a construction group (IT Consultant & Python Engineer) (09/2026 to 10/2026):
Mission for a family-owned construction group of 6 companies in Guadeloupe, carrying out works for the Departmental Council. Their data was stored on a Synology NAS installed by a previous provider, after an incomplete migration from the old Windows server: diverging servers, no file organisation and inconsistent access rights. The managing director expected concrete solutions, not an audit.
➢ **Scoping**: interviews with management, accounting and management control; mission refocused on the data chain actually in use (purchase order → quote → progress statements → Chorus invoicing).
➢ **Audit kit on a USB stick**: portable Python, run on workstations without installing anything and read-only (proven by before / after fingerprints); inventory of ~265,000 files across 5 network shares and an old server (tree structure, Office metadata, duplicates, rights, per-company statistics).
➢ **Migration analysis**: transferred / modified / forgotten report between the old server and the NAS, which revealed writes still going to the old server (payroll, copier scans, invoicing software backups).
➢ **Target folder structure**: 7 variants compared automatically on measured indicators, validated by user tests on a standalone HTML mock-up (simulated rights, predictions archived before the tests) and improved with the accountant's feedback.
➢ **Excel links migration tool**: xlsx / xls / xlsm and analysis of binary formats; inventory, fixes applied to copies with full undo, 78 automated tests; it revealed that migrated workbooks were still silently reading the old server.
➢ **Security takeover**: audit and correction of the NAS SMB / ACL rights (role-based rights), closing the previous provider's access, gradual freeze of the old server.
➢ Step-by-step operating procedures and Word deliverables generated from a style guide (python-docx).
➢ **AI-assisted design**: an agent simulating a user was used to stress-test the folder structure before submitting it to real users.
➢ **Tech environment:** Python 3.12 (portable), pypdf, olefile, python-docx, Synology DSM (SMB, ACL, QuickConnect, Active Backup for Business), Windows Server, Excel (external links), HTML / JavaScript, PowerShell, Claude Code.

5. WONDO - Mobile app for martial arts clubs (08/2026 to present):
iOS / Android app for martial arts clubs in Guadeloupe: members find their schedule, progress and membership in it, and staff run the club from their phone or from a web back office. First club on board: Song Long.
➢ Expo / React Native app with three areas (student, instructor, admin) behind role guards, plus a view and club switcher for the platform administrator.
➢ **Student**: schedule and bookings, membership and plans, progress per discipline (levels, techniques), club news.
➢ **Instructor**: attendance sheet with four statuses (week, day, then class), student search and full student profile.
➢ **Admin**: members and roles, recurring classes (editing a series only changes upcoming sessions), payments recorded from the phone, statistics.
➢ Online registration modelled on the club's paper form (photo from camera or file, summary email, Excel export); approval emails the student their access and activates their membership.
➢ **Generic web back office**: tables and forms generated from a resource registry declared in the API, data partitioned per club.

➢ **FastAPI / PostgreSQL API**: memberships aligned with the school year, Guadeloupe time zone, account and personal data deletion from the app.
➢ **Release**: EAS builds (TestFlight, APK and Google Play), OTA updates, App Store listing under submission; Docker / Traefik deployment through GitHub Actions.
➢ **Tech environment:** React Native 0.81, Expo SDK 54 (Expo Router, EAS Build, EAS Update), TypeScript, TanStack Query, Zustand, React Hook Form + Zod; React, Vite, Tailwind back office, FastAPI, SQLAlchemy, Alembic, PostgreSQL, Docker, Traefik, GitHub Actions.

6. ARC Gestion et Recouvrement - Showcase website, CRM and client portal (08/2026 to 09/2026):
Complete platform for a debt collection firm: showcase website (arc-gr.fr), internal mini-CRM to track collection cases and a private client portal to follow recovered amounts.
➢ Showcase website with an unpaid-invoice impact calculator, SEO and legal pages.
➢ **Mini-CRM**: clients, debt cases, Kanban statuses, reminder history, due-date alerts and search / sorting.
➢ Two-step email reminders, multi-client batch sending, double-send lock.
➢ **Excel / CSV import of client files**: preview before import, column mapping remembered per client, reconciliation and full rollback of an import.
➢ Client report in Word (.docx) with the share due on recovered amounts and an AI-generated comment (Claude API).
➢ **Private client portal**: role-based authentication (JWT), access granted from the CRM, view of outstanding and recovered amounts.
➢ PostgreSQL on an internal Docker network, Alembic migrations run on every deployment, recycle bin (soft delete) and automatic backups.
➢ Hardened VPS (key-based SSH, firewall, fail2ban) and shared Traefik reverse proxy with Let's Encrypt TLS.
➢ **Tech environment:** React, TypeScript, Vite, Tailwind, React Hook Form, Zod, FastAPI, SQLAlchemy, Alembic, python-docx, PostgreSQL, Claude API (Anthropic), Docker, Traefik, GitHub Actions, Linux VPS.

7. Matheva - Private maths tutoring platform (08/2026 to 09/2026):
Three-part platform for a maths teacher (public website, parent area, teacher area), built around a teaching thread: assess, identify needs, support, measure progress and keep parents informed.
➢ Public website with a contact form sent by email (SMTP).
➢ **Public diagnostic test**: question bank, server-side scoring, tiered results and lead capture (“Waiting students” page).
➢ **Teacher area**: students, calendar and session sheets, topic tracking per curriculum, course types and rates.
➢ **Payments module**: each completed or cancelled lesson automatically generates its payment, expected revenue KPI.
➢ **Read-only parent area**: each child's topics, sessions, homework and payments, accessed through a personal link without a password.
➢ Schedule export to Google Calendar (iCal feed).
➢ **Tech environment:** React, TypeScript, Vite, Tailwind, FastAPI, SQLAlchemy, icalendar, PostgreSQL, Docker, Traefik, GitHub Actions.

8. Valado - Migration and security hardening of a Sage Batigest infrastructure (08/2026):
Migration and upgrade of a Sage Batigest infrastructure running on SQL Server, with transfer to a new Windows server and restoration of multi-workstation access.
➢ Migration of Sage Batigest and its SQL Server Express database to a new Windows server, restoration of multi-workstation access.
➢ Configuration of SQL Server, network services and the firewall rules required by Batigest.
➢ Secure remote administration through Tailscale (WireGuard) and RDP, without exposing the RDP service to the Internet, with a dedicated administrator account and access management.
➢ Deployment of RustDesk with unattended access for remote maintenance and user support.
➢ **Server reliability**: automatic service start-up, power management, network access and service continuity.
➢ **Tech environment:** Windows Server, system administration, PowerShell, Sage Batigest, SQL Server Express, Tailscale (WireGuard), RDP, SMB, TCP/IP, Windows firewall, RustDesk.

9. Aivocat - 100% local AI legal assistant (sovereign RAG) (07/2026 to 08/2026):
Freelance mission for a law firm: lawyers cannot send their clients' documents to ChatGPT. Designed and delivered, single-handedly, an AI assistant that runs entirely on one of the firm's PCs (an RTX 3060 is enough), with no Internet connection: no data ever leaves the machine (professional secrecy).
➢ **Complete RAG pipeline**: multi-format ingestion (PDF, Word, Excel, emails) with OCR, chunking, embedding, hybrid search (BM25 and vectors) with cross-encoder reranking.
➢ Answers that cite the exact passages they rely on, or decline to answer when nothing supports them (abstention thresholds calibrated on real measurements).
➢ Integration of statutory texts (Civil Code, 1989 law…) from the official LEGI/DILA open data, split article by article and queried alongside the case documents.
➢ **Business features**: case summary sheet, chat, legal analysis, SWOT and attack or rebuttal strategies.
➢ Case files read from a USB stick in read-only mode, the assistant's memory kept on a separate stick.
➢ **VRAM management on 12 GB**: embeddings, reranker and 9B LLM take turns sharing the GPU.
➢ **Industrialisation**: fully offline installation package (14 GB) with SHA-256 checksum verification, compiled code (Nuitka), machine-bound Ed25519-signed licence, diagnostic and backup scripts.
➢ **Results:**
- About 650 automated tests for about 15,000 lines of Python.
- Automated evaluation "jury" scoring citation accuracy and the rate of correct answers on test case files, including a fictitious tax case generated for the purpose.

➢ **Tech environment:** Python 3.12, FastAPI, SQLite, HTML/JS, Ollama, Qwen 3.5 9B, bge-m3, bge-reranker-v2-m3, Qdrant (base vectorielle), RapidOCR, Nuitka, PowerShell.

Tech Lead, Software Architect & Lead Developer - DREVIO Since 05/2026

Paid client mission – SaaS solution for the automotive sector

Company context:
Design of DREVIO, a SaaS platform that digitises and automates vehicle damage assessment: a case is created from a vehicle and photographs, the damage is analysed by artificial intelligence, then a first estimate usable by body-shop professionals is generated. Phase 1 completed in August 2026, mobile app in progress. Phase 2 started in September 2026 with HellBoy (training dataset), full launch in November 2026.

Key projects:
1. DREVIO Phase 2 - HellBoy, training dataset platform (09/2026 to present):
To train a future model that recognises bodywork damage and estimates repair costs, DREVIO needs photos tied to the price actually invoiced. HellBoy produces this raw material: every validated case (photos, located damage, invoiced price) becomes a training sample.
➢ Import of the archives sent by body shops (ZIP: photos and repair estimate as PDF): reading of the vehicle, hourly rates, parts and operations, with a check that the rebuilt total matches the printed pre-tax total to the cent.
➢ Closed vocabulary (32 body zones, 13 damage types, 3 severities) in a domain package shared by the API and the dashboard, enforced in the database through foreign keys.
➢ **Photo and annotation workshop**: damage entered directly on the photo, bounding boxes normalised in the file's coordinate system, zones that can be moved and resized.
➢ Review workflow (draft, to review, validated or rejected), frozen validated cases, role-based permissions (admin, reviewer, annotator, viewer) enforced by the API.
➢ Read-only import of DREVIO cases whose human assessment is validated: only the human costing is taken, so the model does not learn to imitate the current AI.
➢ Versioned dataset export (validated cases only, amounts in integer cents, price context).
➢ Docker deployment on DREVIO's VPS behind Traefik, with its own database and subdomain.
➢ **Tech environment:** TypeScript, Node.js, Fastify (npm workspaces monorepo), React 19, Vite, TanStack Query, Zustand, PostgreSQL, Docker, Traefik, GitHub Actions.

2. DREVIO Mobile - Tech Lead of the iOS / Android app (06/2026 to present):
Mobile app that estimates vehicle damage from photos analysed by AI: it produces an assessment report, calculates a quote for each partner garage and handles online payment.
➢ **Project set-up**: app initialisation, development and production environments, choice of the stack and of a feature-based architecture (auth, scan, vehicle, payment, notifications).
➢ **Mentoring a junior developer**: split of the product into more than 60 deliverable work packages,

writing and prioritising tasks, tracking their delivery.
➢ **Quality and code review**: review of more than 140 pull requests, hardening after review, TypeScript and lint checks.
➢ **Business logic**: design of the per-garage price calculation, which applies each garage's own rates and pricing mode.
➢ Upgrade of the app from Expo SDK 54 to SDK 57.
➢ Features delivered under my lead:
- Guided camera scan with photo quality checks and real-time tracking of the AI analysis.
- AI report with damage location, analysis re-run and request for a human assessment.
- Garage search by geolocation, with sorting and filters.
- Stripe payment (deposit, balance, refund) through Supabase Edge Functions.
- Insurance and claim flow, management of leased vehicles (long-term lease / lease with purchase option).
- Push notifications (Firebase), Google OAuth sign-in and OTA updates.

➢ **Tech environment:** React Native 0.86, Expo SDK 57 (Expo Router), TypeScript, Zustand, TanStack Query, React Hook Form + Zod, Supabase (Edge Functions), Stripe, Firebase FCM.

3. DREVIO Phase 1 - SaaS platform for vehicle damage assessment (05/2026 to 08/2026):
➢ **Distributed architecture**: React Native/Expo mobile app, React dashboard, .NET 10 API and asynchronous analysis worker.
➢ **Vertical Slice API**: authentication, vehicles, cases, media, damage, estimates, payments and event history.
➢ Multimodal analysis pipeline combining photographs, vehicle data, OpenAI models and structured validation of the results.
➢ PostgreSQL database design and security with Supabase Auth, JWT, Row Level Security and private photo storage.
➢ Study and preparation of an API integration with Lacour/IRIS to obtain repair-costing data recognised by automotive professionals.
➢ Dockerisation and VPS deployment with Traefik, GitHub Actions and GitHub Container Registry.
➢ Definition of the MVP scope, backlog and technical architecture; coordination of development within a two-person team.
➢ **Tech environment:** C#, .NET 10, ASP.NET Core, Entity Framework Core, React, React Native, Expo, TypeScript, TanStack Query, PostgreSQL, Supabase, OpenAI API, Docker, Traefik, GitHub Actions, Linux.

FullStack Software Engineer .NET / WPF — 03/2025 to 03/2026
PROELAN / Schneider Electric

Company context:
Part of the "CEART" Agile Release Train (ART) of 90+ people, in the "Topology Manager" team (5 people) following the SAFe methodology. Architecture defined upstream by Schneider Electric: contribution to feature development and migration.

Key projects:
1. Topology Manager project (desktop IDE - C# / WPF):
➢ FullStack development of **Topology Manager**, an IDE in **C# / WPF** enabling real-time collaborative programming of PLCs (API in .NET 8, WPF desktop client).
➢ **Migration and extension** of the feature catalogue from the legacy application **Control Expert Classic** (C++) to the new client-server architecture of TM.
➢ **Technical challenges:**
- **Legacy codebase integration**: interoperability and adaptation of the C++ features of Control Expert Classic within the modern .NET ecosystem.
- **Performance troubleshooting**: investigation and fixing of the root causes of slowdowns in a mature application (10+ years old).
- **Quality assurance**: unit and functional tests implemented for every new feature.

➢ **Tech environment:** C#, WPF, .NET 8, Fody framework, unit tests, C++ (legacy).

2. SET project (Schneider Electric Toolkit - Web):
➢ Contribution to the **SET unified web platform** bringing together access to all Schneider products

(Control Expert, Machine Expert, Automation Expert).
➢ **Front-end development** to integrate **Control Expert Classic** as a **Micro-Frontend** in the SET interface.
➢ **New front-end features** implemented, such as the integration of a Git client using **Angular** and **WebKit**.
➢ **Collaboration** with the back-end teams on the existing API (shared with TM) to define front-end needs.

➢ **Tech environment:** Angular, TypeScript, WebKit, Micro-Frontends, API REST .NET.

Key skills demonstrated:

- **Adaptability**: work across a broad technology spectrum, from **desktop WPF** (.NET 8) to **modern web front-end** (Angular/TypeScript).
- **Work on complex legacy systems**: ability to navigate, understand and evolve an old, critical codebase.
- **International collaboration**: initial training by the Spanish team, then collaboration with Romanian developers in a distributed team.

AWS Cloud Architect and Python Engineer 06/2024 to 02/2025
Ayming

Company context:
Ayming is an international consulting firm specialised in improving companies' operational and financial performance. With more than 35 years of experience, it supports its clients in areas such as innovation, taxation and digital transformation. As a software engineer, I contributed to several strategic projects automating business processes and securely managing sensitive data.

Key projects:

1. SYLAE (scraping bot):

- Goal: automate the retrieval of the amounts and details of financial aid for work-study students through the SYLAE platform.
- Technical delivery:
  - Switch from Playwright to Requests after analysing the client-server requests, reducing the scraping run time.
  - Extraction of the JSON responses, structured storage in DynamoDB, and generation of Excel files with Pandas for the end client. Each row contained key information with cross-references to the downloaded PDFs.
  - Integration of AWS XRay to provide precise logs and raise alerts when a problem occurs.

2. ATMP (NET ENTREPRISE scraping):

- Goal: automate the retrieval of AT/MP (workplace accident) rates, calculation sheets and certificates for about 5,000 client accounts through NET ENTREPRISE. ATMP already existed, in Java, monolithic and built on dated paradigms: Ayming asked me to rebuild it as multiple services, adapted to AWS and secured.
- Technical challenges:
  - Identification of the cookies needed to keep a session alive and send the server requests.
  - Fine-grained handling of specific errors (e.g. terms of use to accept, expired password), with automatic update of the account status in DynamoDB and notification through an automated mailing system.
- Scalable architecture:
  - Designed the entire target architecture on AWS, presented to the team as diagrams (Miro), then implemented an asynchronous model on AWS ECS where each task runs independently and triggers a dedicated SnapLogic pipeline to store the data and log the results.

3. RobotCredentials (credential management):

- Goal: provide a secure solution to manage and access the credentials required by the scraping processes.
- Technical delivery:
  - Creation of an RDS database storing client accounts with AES-256 encrypted passwords.
  - Development of an ASP.NET API in Clean Architecture allowing SnapLogic and the scrapers to access the credentials securely.
  - Implementation of a secure encryption/decryption system guaranteeing the confidentiality of sensitive data.

Tasks & achievements:

Development & technical optimisation:

- Creation of high-performance Python scrapers with Requests, optimised with safe caching to avoid redundant processing.
- Robust handling of specific errors (e.g. account status updated in DynamoDB depending on the error detected).

Infrastructure & security:

- Design of a secure solution with RDS to store client credentials.
- Development of an ASP.NET API with secure AES encryption/decryption.
- Deployment on AWS ECS with SnapLogic integration to orchestrate the workflows.

Scalable architecture:

- Implementation of an asynchronous model where ECS tasks run independently and call back a dedicated SnapLogic pipeline.
- Use of AWS ECS best practices to ensure efficient horizontal scaling (load-based auto-scaling).

Monitoring & reporting:

- Integration of AWS XRay into the Python code to provide precise logs available to the infrastructure.
- Automated mailing service summarising the scraping results and reporting any problem detected.

Documentation & collaboration:

- Complete technical documentation in Markdown on Azure DevOps.
- Explanatory diagrams of the overall architecture (RDS, DynamoDB, S3, SnapLogic).
- Regular presentations to stakeholders in dedicated meetings.

Results & impact:

1. **SYLAE:** full automation of a manual process that took a dedicated employee several weeks per month, saving significant time and drastically reducing human errors.
2. **ATMP:** successful move to a modern, cost-effective Python architecture, cutting Java-related costs while increasing the system's flexibility and reliability. Critical project: according to the IT director, each week of delay meant a loss of about €1.5 million.
3. **RobotCredentials:** secure solution guaranteeing the confidentiality of credentials while making them easy to access for SnapLogic and the scrapers.

Tech environment:

- **Languages:** Python, Java, C#.
- **Technologies & tools:** Playwright, Requests, Pandas, Docker, AWS ECS, DynamoDB, S3, RDS (PostgreSQL), AES-256, AWS XRay, SnapLogic.
- **Frameworks & patterns:** ASP.NET Clean Architecture (API), secure AES encryption/decryption, CI/CD pipelines

C# Software Engineer and Certified SAFe Scrum Master — June 2023 to May 2024

AViSTO / Schneider Electric

Company context:

- Scrum Master of the 8-member Device Integration team (Product Owner, developers, testers).
- Part of an international Agile Release Train (ART) of 90 people across France, Germany, Serbia, India and Singapore. Product architecture defined by Schneider Electric.
- Daily collaboration in English to coordinate teams and deliver complex software solutions.

Project context:

- Development of an IDE in C# WinForms based on CoDeSys to program Schneider Electric industrial PLCs.
- Business process automation through TypeScript and Python scripts generating configuration files.
- Integration of an Angular application into a .NET WinForms interface, improving interoperability between existing systems.
- Work on the EdgeIO product, recently unveiled publicly, which manages various industrial modules with different communication protocols (Ethernet, ModBus, Sercos).

Tasks & achievements:

Software Engineer (.NET/Python):

- In just 6 months, with two colleagues also new to the team, delivered work equivalent to what the previous team had achieved in two years.
- Integration of a new Angular application into the Machine Expert interface (.NET WinForms), adapting the features developed by the Angular front-end team to the more complex needs of Machine Expert.
    - The Angular front-end team only handled the features of the EdgeIO module on a dedicated IP, whereas Machine Expert had to support several module types with various protocols (Ethernet, ModBus, Sercos).
- Verification and fixing of the communication between the software and the hardware tools through the OPCUA and ModbusTCP industrial protocols.
- Development and improvement of the general PLC (programmable logic controller) management logic in the Machine Expert application.
- In-depth functional testing and delivery of new software versions with an optimised release cycle.

Scrum Master (SAFe certified):

- Organisation and facilitation of Agile ceremonies: System Demo, PI Planning, Daily Scrum, SoS (Scrum of Scrums).
- Coordination with teams outside the ART (Firmware, Software) to ensure smooth communication between stakeholders.
- Proactive handling of the team's impediments and solutions put in place to speed up development.
- Writing and rigorous tracking of Jira tickets to meet deadlines and objectives.

Tech environment:

- **Languages:** C#, Python, TypeScript.
- **Technologies:** WinForms, Angular, OPCUA, ModbusTCP, Sercos.
- **Tools:** Jira, GitHub Enterprise, iObeya.
- **Agile methodologies:** SAFe Framework, Scrum.

FullStack Engineer C#/Python 10/2019 – 06/2023
DIGITOM

Company context:

- Member of the ERP unit (team of 2 to 3 people).
- Odoo ERP integration or custom application development.
- Collaboration with the BI units and the group's departments.

BusinessZone (internal),
.NET application for displaying BI reports – September 2020 to May 2023

Project context:

- An ASP.NET MVC web application in C# with Clean Architecture enabling controlled sharing of Power BI reports with DIGITOM's clients within their departments.
- Tech Lead on the whole project, overseen by Régis GEROMEGNACE
- Need identified from several projects and upstream issues,
- Clean Architecture in place for a modular, maintainable and scalable application.
- Controlled sharing of Power BI reports:
  - Authentication and authorisation, role management and role overrides
  - Power BI integration, controlled and administered sharing
  - Data security (sensitive accounting, salary or revenue data)
- Planned features:
  - Multi Tenant: access from the same instance to several Azure AD/Power BI Service instances
  - Pre-filter: for a shared Power BI report, dynamic filtering based on the BZ login
  - Multi DB: reports that can adapt to most ERP databases on the market.
  - Multi Service: standardised reports covering the main needs of any company. Continuous improvement of the reports of clients already subscribed to BZ
  - Multi BI: support for BusinessObject, Tableau or even visualisation frameworks (JS, Python, R or others) to avoid limiting the solution to Power BI Embedded.

Tasks & achievements:

- **Tech Lead** on the whole project
  - Idea and specifications
  - UML design
  - POC presentation to management
  - Product launch
- **Data Analyst** SQL/Python:
  - Gathering the main client needs (intermediate management balances, commissions, KPIs)
  - Creation of the required ETLs
  - Report design
- **FullStack** .NET Engineer:
  - .NET Clean Architecture set-up
  - Power BI integration
  - MVC API set-up
  - ASP.NET front-end design and implementation

Tech environment:

- **Languages & frameworks:** C#, ASP.NET MVC, Entity Framework.
- **BI & Cloud:** Power BI Service/Embedded, Azure AD.
- **Database**: SQL Server.
- **Principles & patterns:** SOLID, CQRS, Mediator, Repository Pattern.
- **Additional tools:** AutoMapper, MediatR, Fluent Validation, Swagger.

Odoo Docker (internal)
Containerising an ERP application – 2 months

Project context:

- Creation of an Odoo container to avoid using virtual machines
  - Container developed to make onboarding easier for the company's future developers.
  - Support for official Odoo versions and adaptation to older versions
  - Makes it possible to run local instances of all our production clients.
  - Project linked to **AutoSauvegarde**:
    - Odoo module created to back up client instances in case of incident.

By loading any production backup, the container therefore allowed very fast development, testing or repair on a development instance.

Tasks & achievements:

- DevOps / Python Engineer:
  - Custom Odoo Dockerfile and additional scripts
  - Tests with client instances
  - Documentation for newcomers
  - Settings to configure instances (client or blank, database, static files)

Tech environment:

- **Docker, Dockerfile, Docker Compose**
- **Python, Venv**
- **PostgreSQL**

PowerBI on Odoo (internal project), standard Power BI reports for Odoo – 2 months

Project context:

- Use of Power BI on Odoo's PostgreSQL databases for performance analysis
- Fills the gaps of Odoo's Dashboard module
- DIGITOM, Karukero / SUEZ, Pk Trading, Groupe Barboteau

The project then made it possible to build a good number of reports to integrate into **Business Zone**.

Tasks & achievements:

- **Data Analyst** SQL/Python:
  - Gathering the main client needs (intermediate management balances, commissions, KPIs)
  - Creation of the required ETLs
  - Design of various performance indicators
    - Revenue
      - By department, period and client,
    - Intermediate management balances
    - Water distribution
    - Leak repairs
    - Commission calculation
      - By salesperson, period, client, product category
    - Average basket
  - Report design

Tech environment:

- **Python**
- **Odoo**
- **PostgreSQL**
- **Docker Desktop**

Power BI Coaching, training of Groupe Barboteau managers – 3 months – October 2020 to December

Project context:

- Training of the managers of a large retail group in using Power BI to manage their data:
  - The group had a strong need for ETL and data collection because of its many companies running different management software.
  - The training aimed to show how to use Power BI to extract relevant information for decision-making.
  - Learning the DAX language and building reports. 1 day of basics, 2 days of report building, and 2 days of work based on the participants' specific requests

Tasks & achievements:

- Trainer 4 days out of 5:
  - Basic features (data transformation, relationships, visuals)
  - Advanced features (Power Query, DAX, sharing)
  - Guided design of 2 reports on their data (Pareto principle, mass retail)

Tech environment:

- **Power BI Desktop**
- **ETL**
- **DAX, PowerQuery**
- **Excel**

Power BI Support, automated report updates – 6 months – January 2021 to June 2021

Project context:

- Automated update of several Power BI reports when a CSV file is received at a specific email address:
  - The file is saved to an Azure Blob, then to an Azure SQL database to be loaded into Power BI and update all client reports.
  - C2i's support team uses the Autotask tool to manage its client interventions.
  - Automation solution proposed to tackle the growing number of clients and the manual workload.

Project linked to the **Business Zone.**

Tasks & achievements:

- BI Analyst / DevOps Engineer:
  - Solution design and presentation
  - Pipeline set-up
    - Power Automate management
    - Azure Blob Storage management
    - Azure SQL Server integration
  - Update and adaptation of the base report
  - End-client-focused report generation and updates with Power BI Service

Tech environment:

- **Power BI Service**
- **Azure Blob Storage**
- **Azure SQL Database**
- **Autotask**

Gestrav, "Urgence Eau Guadeloupe" 2020 mission: coordinating water leak repairs – 5 months – June 2020 to October 2020

Project context:

- 2020 "Urgence Eau Guadeloupe" mission: facing water shortages, the State ordered prefectoral requisitions entrusted to the Karukér'Ô / SUEZ consortium. Within a team of 3 people, built a management tool coordinating the leak detectors, the repair teams and the network operator (SIAEAG):
- Geolocation set-up and presentation of the data once retrieved, with optimised routes.
- Development in C#, ASP.NET, JavaScript and Power BI integration.
- Power BI data and reports on the mission's indicators (leaks detected and repaired), followed in real time by the public on the urgence-eau-guadeloupe.org website.
- On-site work at Eau d'Excellence to integrate the leak data and develop the leak geolocation interface of the Gestrav application.
- Gestrav became the official SMGEAG application for detecting leaks and tracking their repair.

Tasks & achievements:

- Power BI geolocation POC
- Power BI data and reports on the indicators published on urgence-eau-guadeloupe.org
- Development of part of the Gestrav user interface
- OpenStreetMap integration in the .NET application
- Design and implementation of the map features (routes, display of points and associated data)

Tech environment:

- **C#, ASP.NET**
- **Javascript**
- **Power BI**
- **Excel**
- **OpenStreetMap**

Degrees

2019 - Bachelor of Information Technology (MIAGE) - Université des Antilles (Guadeloupe, France)
2022 - Master of Information Technology (MIAGE) with a specialization in Data Science - Université des Antilles (Guadeloupe, France)
2023 - CS50x and CS50AI certification - Harvard (Cambridge, Massachusetts)
2024 - SAFe 6.0 Scrum Master certification – Scaled Agile Framework Inc.

Competitions

Google HashCode	2020: **5830**th / 10,724 2021: **1545**th / 9,004 2022: **4728**th / 9,031
Hack my Flat	June 2022: **1st** place IAD company hackathon